NATHACKS 2026 · EDMONTON

PathSense User Stories

53 user stories · 187 story points · 6 people · 4 phases

Version 2.0 · November 7, 2026 · natHACKS 2026. Every story is a GitHub issue with acceptance criteria and live status; click an ID to open it. Full details are in PLAN.md.

1. Who Owns What

Story points per person per phase. One point is about one focused hour; each person plans for about 8 points per phase.

Person	GitHub	Phase 1	Phase 2	Phase 3	Phase 4	Stories	Points
Tatsat	@tatsatshah05	8	7	8	8	9	31
Ipsa	@ipsamanhas	8	8	8	8	10	32
Estella	@estella1221	8	8	8	8	9	32
Kaleb	@kalebpham	8	8	8	8	9	32
Kashish	@kashishgupta2911	7	8	7	7	8	29
Matthew	Not yet a collaborator	8	8	8	7	8	31
Team		**47**	**47**	**47**	**46**	**53**	**187**

Phases: **1. Data Foundation** (hours 0–10) · **2. Vision and Routing** (hours 10–22) · **3. Web App and Alerts** (hours 22–34) · **4. Production and Launch** (hours 34–48).

2. All User Stories

Phase 1: Data Foundation (hours 0–10)

Checkpoint: Live items from every Must source appear in `GET /api/obstructions`; `GET /api/status` shows each source's freshness; the city walking network loads within its memory budget; the API and web app deploy from `main` with CI passing; the 511 Alberta camera list is counted.

ID	User story	Owner	Points	Priority
PS-101	**Obstruction contract and store** As a developer, I want one shared format for every obstruction, so that each data source plugs in the same way and the app never depends on where an item came from.	Tatsat	3	Must
PS-102	**Deployable skeleton with CI** As the team, we want the API and web app deployed from day one with automatic checks, so that every merge is tested and the live site never breaks.	Matthew	5	Must
PS-103	**Source scheduler and status page** As a user, I want to see how fresh each data source is, so that I can trust what the app tells me.	Matthew	3	Must
PS-104	**City walking network** As a pedestrian anywhere in Edmonton, I want the app to know every walkable path, including stairs, crossings and elevators, so that routes match how I move.	Tatsat	5	Must

ID	User story	Owner	Points	Priority
PS-105	**LRT elevator and escalator adapter** As a wheelchair user, I want to know when an elevator at an LRT station is out of service, so that I am never sent to a station I cannot use.	Kaleb	3	Must
PS-106	**Map LRT elevators to the walking network** As a wheelchair user, I want a broken elevator to change my route, not just show a warning, so that the route I get is truly step-free.	Kaleb	3	Must
PS-107	**Construction sidewalk closures adapter** As a pedestrian, I want sidewalk closures from construction permits on my route, so that I do not walk into a closed sidewalk.	Kaleb	2	Must
PS-108	**Traffic disruptions adapter** As a pedestrian, I want to know about road closures and incidents along my route, so that I can expect detours at crossings.	Ipsa	2	Must
PS-109	**Weather and black-ice watch** As someone walking in winter, I want a warning when the weather makes black ice likely, so that I can choose a safer route or time.	Estella	3	Must
PS-110	**311 snow complaints and trail closures** As a pedestrian, I want to see where neighbours report snowy sidewalks and where trails are closed, so that I know which areas to avoid.	Estella	3	Must
PS-111	**Snow-clearing status adapter** As a wheelchair user, I want to know when a plow has just passed a road on my route, so that I am warned about snow ridges across curb ramps.	Estella	2	Should
PS-112	**Public camera access** As the vision team, we want legal, reliable access to public road cameras, so that the model can read winter conditions without our own hardware.	Ipsa	3	Must
PS-113	**Web app shell and live map** As a user, I want a fast, accessible map that shows every current obstruction with icons and labels, so that I can see the city's conditions at a glance.	Kashish	5	Must
PS-114	**Wireframes for the main screens** As a user, I want the most important information first on every screen, so that I can decide quickly and safely.	Kashish	2	Must
PS-115	**Label guide and labelling tool** As a labeller, I want a clear label guide and a fast keyboard tool, so that I can label hundreds of camera images consistently.	Ipsa	3	Must

Phase 2: Vision and Routing (hours 10–22)

Checkpoint: The model meets its target on cameras it never saw in training; camera readings and the black-ice watch reach the obstruction feed; `GET /api/route` returns routes with alternates and a full report; a test elevator outage reroutes a wheelchair route.

ID	User story	Owner	Points	Priority
PS-201	**Collect and label training data** As the vision team, we want a labelled set of real camera images across day, night, dry, wet, snow and ice, so that the model learns the conditions it will see.	Estella	5	Must
PS-202	**Train the road-condition model** As a user, I want the app to recognise snow and ice from camera images, so that warnings come from what cameras see, not only from the forecast.	Ipsa	5	Must
PS-203	**Evaluate the model and write the model card** As a user and a judge, I want honest numbers on how well the model works, so that I know when to trust it.	Ipsa	3	Must
PS-204	**Camera vision worker** As a user, I want camera readings refreshed automatically, so that the map shows what the road looks like now.	Kaleb	5	Must
PS-205	**Black-ice risk from cameras and weather** As someone walking in winter, I want black-ice warnings that combine what cameras see with the temperature, so that warnings are specific to a place and time.	Kaleb	3	Must
PS-206	**Match obstructions to the network** As a user, I want every obstruction tied to the exact paths it affects, so that routes avoid the right streets.	Matthew	5	Must
PS-207	**Profile routing with alternates** As a wheelchair, walker, low-vision or standard user, I want the best route for how I move plus up to two alternates with distance and time, so that I can choose.	Tatsat	5	Must
PS-208	**Route report** As a user, I want a list of every obstruction on or near my route, grouped by indoor, sidewalks, roads and weather, so that I know exactly what I will face.	Matthew	3	Must
PS-209	**Google Maps and Apple Maps hand-off** As a user, I want to open my route in Google Maps or Apple Maps, so that I get turn-by-turn directions in an app I already know.	Tatsat	2	Must
PS-210	**Planner screens with mock data** As a user, I want to enter where I am and where I am going, choose how I move, and compare routes, so that planning takes seconds.	Kashish	5	Must
PS-211	**Route report screen** As a screen-reader user, I want the route report as a clear text list, so that the map is never the only way to get information.	Kashish	3	Must
PS-212	**Accessibility review of designs** As a disabled user, I want accessibility designed in from the start, so that the app works for me on day one.	Estella	3	Must

Phase 3: Web App and Alerts (hours 22–34)

Checkpoint: On a phone, the deployed site plans a route anywhere in Edmonton, shows the full report, updates live when an obstruction changes, sends a push alert for a saved route, and opens the route in Google Maps.

ID	User story	Owner	Points	Priority
PS-301	**Connect the planner to the live API** As a user, I want real routes and reports on my phone, so that I can rely on the app for a trip.	Kashish	5	Must
PS-302	**Address and place search** As a user, I want to type an address or a place name, so that I do not have to find it on the map.	Matthew	3	Must
PS-303	**Live updates** As a user with a route open, I want it to update by itself when something changes, so that I learn about a new obstruction before I reach it.	Matthew	5	Must
PS-304	**Saved routes and push alerts (server)** As a commuter, I want an alert when my usual route becomes blocked or much slower, so that I can leave early or go another way.	Tatsat	5	Must
PS-305	**Replay mode** As the demo team, we want to replay recorded real data from every source, so that the demo works when the weather does not cooperate, and judges always know it is a replay.	Tatsat	3	Must
PS-306	**Alerts page and push client** As a commuter, I want to save a route and turn on alerts in two taps, so that I am told when it changes.	Estella	5	Must
PS-307	**Accessibility settings** As a low-vision user, I want to control speech, text size and motion, so that the app works the way I need.	Estella	3	Must
PS-308	**Community reports with photo check** As a pedestrian, I want to report snow, ice or a blocked path in a few taps, with an optional photo, so that others are warned about places no camera can see.	Ipsa	5	Should
PS-309	**Camera readings on the map** As a user, I want to see each public camera's reading and when it was taken, so that I understand where an ice warning comes from.	Ipsa	3	Must
PS-310	**City dashboard** As a City operations coordinator, I want one list of every barrier ranked by how many trips it affects, so that crews clear the most important places first.	Kaleb	3	Should
PS-311	**Walk and LRT trips** As a wheelchair user who rides the LRT, I want trips that combine walking and the LRT and only use stations whose elevators work, so that I can cross the city step-free.	Kaleb	5	Should
PS-312	**About, sources and privacy pages** As a user, I want to know where the data comes from and what the app stores about me, so that I can trust it.	Kashish	2	Must

Phase 4: Production and Launch (hours 34–48)

Checkpoint: The release checklist in PLAN.md is complete, `v1.0` is tagged, the demo is rehearsed five times, and the project is submitted by hour 46.

ID	User story	Owner	Points	Priority
PS-401	**Security and privacy review** As a user, I want my location and reports handled safely, so that I can trust the app.	Tatsat	3	Must
PS-402	**Monitoring and runbook** As the team on demo day, we want to know within a minute if anything breaks and how to fix it, so that the live demo never fails.	Tatsat	3	Must
PS-403	**Release v1.0** As the team, we want a clean, tagged release at the feature freeze, so that we demo and submit exactly what we tested.	Tatsat	2	Must
PS-404	**Performance and load** As a user on a phone network, I want routes in about a second, so that the app feels instant.	Matthew	5	Must
PS-405	**Data backup and restore** As the team, we want backups we have actually restored, so that a database problem cannot end the demo.	Matthew	2	Must
PS-406	**End-to-end tests** As the team, we want the critical path tested automatically on every merge, so that late fixes never break the demo.	Kaleb	5	Must
PS-407	**Data quality checks** As a user, I want the app to notice when a city feed changes or breaks, so that I am never shown silently wrong data.	Kaleb	3	Must
PS-408	**Accessibility audit** As a disabled user, I want the whole app to meet WCAG 2.2 AA, so that an accessibility app is itself accessible.	Estella	5	Must
PS-409	**Pitch deck and Devpost** As a judge, I want a clear, honest story backed by real numbers, so that I can trust what the team built.	Estella	3	Must
PS-410	**Web polish and performance** As a judge opening the site from a QR code, I want it to load fast and look finished, so that the first impression is strong.	Kashish	5	Must
PS-411	**Judge page and launch assets** As a judge, I want to open the app on my own phone in seconds, so that I can try it during judging.	Kashish	2	Must
PS-412	**Model monitoring and fallback** As a user, I want ice warnings to keep working when cameras or the model fail, so that I am never left without a warning.	Ipsa	3	Must
PS-413	**Technical write-up** As a judge or future contributor, I want a clear technical overview, so that I can understand and extend the system.	Ipsa	3	Must
PS-414	**Backup demo video** As the team, we want a recorded demo of the full product, so that judging goes ahead even if the network fails.	Ipsa	2	Must

3. Stories by Person

Tatsat (@tatsatshah05): 31 points

Phase	ID	Story	Points
1. Data Foundation	PS-101	Obstruction contract and store	3
1. Data Foundation	PS-104	City walking network	5
2. Vision and Routing	PS-207	Profile routing with alternates	5
2. Vision and Routing	PS-209	Google Maps and Apple Maps hand-off	2
3. Web App and Alerts	PS-304	Saved routes and push alerts (server)	5
3. Web App and Alerts	PS-305	Replay mode	3
4. Production and Launch	PS-401	Security and privacy review	3
4. Production and Launch	PS-402	Monitoring and runbook	3
4. Production and Launch	PS-403	Release v1.0	2

Ipsa (@ipsamanhas): 32 points

Phase	ID	Story	Points
1. Data Foundation	PS-108	Traffic disruptions adapter	2
1. Data Foundation	PS-112	Public camera access	3
1. Data Foundation	PS-115	Label guide and labelling tool	3
2. Vision and Routing	PS-202	Train the road-condition model	5
2. Vision and Routing	PS-203	Evaluate the model and write the model card	3
3. Web App and Alerts	PS-308	Community reports with photo check	5
3. Web App and Alerts	PS-309	Camera readings on the map	3
4. Production and Launch	PS-412	Model monitoring and fallback	3
4. Production and Launch	PS-413	Technical write-up	3
4. Production and Launch	PS-414	Backup demo video	2

Estella (@estella1221): 32 points

Phase	ID	Story	Points
1. Data Foundation	PS-109	Weather and black-ice watch	3
1. Data Foundation	PS-110	311 snow complaints and trail closures	3
1. Data Foundation	PS-111	Snow-clearing status adapter	2
2. Vision and Routing	PS-201	Collect and label training data	5
2. Vision and Routing	PS-212	Accessibility review of designs	3
3. Web App and Alerts	PS-306	Alerts page and push client	5
3. Web App and Alerts	PS-307	Accessibility settings	3
4. Production and Launch	PS-408	Accessibility audit	5
4. Production and Launch	PS-409	Pitch deck and Devpost	3

Kaleb (@kalebpham): 32 points

Phase	ID	Story	Points
1. Data Foundation	PS-105	LRT elevator and escalator adapter	3
1. Data Foundation	PS-106	Map LRT elevators to the walking network	3
1. Data Foundation	PS-107	Construction sidewalk closures adapter	2
2. Vision and Routing	PS-204	Camera vision worker	5
2. Vision and Routing	PS-205	Black-ice risk from cameras and weather	3
3. Web App and Alerts	PS-310	City dashboard	3
3. Web App and Alerts	PS-311	Walk and LRT trips	5
4. Production and Launch	PS-406	End-to-end tests	5
4. Production and Launch	PS-407	Data quality checks	3

Kashish (@kashishgupta2911): 29 points

Phase	ID	Story	Points
1. Data Foundation	PS-113	Web app shell and live map	5
1. Data Foundation	PS-114	Wireframes for the main screens	2
2. Vision and Routing	PS-210	Planner screens with mock data	5
2. Vision and Routing	PS-211	Route report screen	3
3. Web App and Alerts	PS-301	Connect the planner to the live API	5
3. Web App and Alerts	PS-312	About, sources and privacy pages	2
4. Production and Launch	PS-410	Web polish and performance	5
4. Production and Launch	PS-411	Judge page and launch assets	2

Matthew (GitHub invite pending): 31 points

Phase	ID	Story	Points
1. Data Foundation	PS-102	Deployable skeleton with CI	5
1. Data Foundation	PS-103	Source scheduler and status page	3
2. Vision and Routing	PS-206	Match obstructions to the network	5
2. Vision and Routing	PS-208	Route report	3
3. Web App and Alerts	PS-302	Address and place search	3
3. Web App and Alerts	PS-303	Live updates	5
4. Production and Launch	PS-404	Performance and load	5
4. Production and Launch	PS-405	Data backup and restore	2

4. Story Point Scale

Points	Size	Example
1	Trivial	Add a field to an existing endpoint
2	Small	One simple data adapter with tests
3	Medium	An adapter with special rules, or a small screen
5	Large	The routing engine, model training, or the planner screens
8	Too large	Split it into smaller stories